Milestone 2 Data Vizualization

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Sketch Website

1 Tools

For this project, we will utilize several tools to implement our visualizations. Specifically, we will rely on **Three.js** for 3D rendering, **Vite** as our development and build tool, and **Node.js** for managing the backend and dependencies. The GitHub REST API will serve as our core data source, allowing us to fetch and process all necessary repository information.

2 Useful Lectures

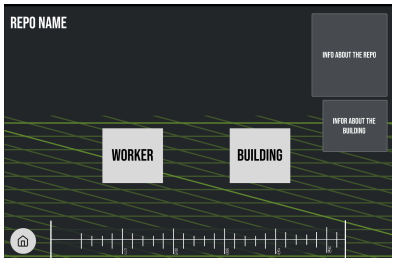
In terms of required knowledge, we will build upon concepts introduced in lectures. Lectures 1 to 4 provide the necessary foundation in JavaScript, which is essential for implementing the project. Lectures 6 and 7 will guide our understanding of visualization principles. Additionally, the storytelling lecture will help us design a more engaging and user-centered experience.

3 Main goals

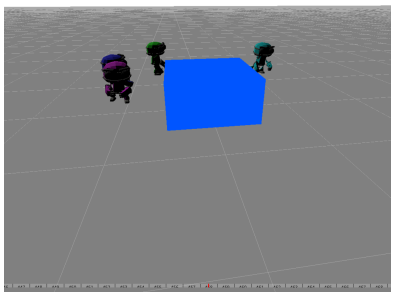
The objective of this project is to visualize the evolution of a Git repository over time in an intuitive and engaging way. The visualization will allow users to explore how a repository grows, who contributes to it, and how activity changes throughout its history. The user will begin by providing the URL of a Git repository (1a). Based on this input, the system will retrieve and process relevant repository data to generate the visualization.

Our core concept represents the repository as a building, where its development over time is reflected through its structure and activity (1c). Contributors to the repository are depicted as workers who actively participate in constructing the building. These workers will appear dynamically, performing actions (e.g., "hammering") that symbolize their contributions. Each worker will be interactive, allowing users to click on them to access detailed contributor information, such as their name, commit count, and activity level (Figure 2a). As time progresses, the visualization will update to reflect changes in the repository, such as new contributions or increased activity. A timeline interface will be displayed at the bottom of the screen.

(a) Home Page

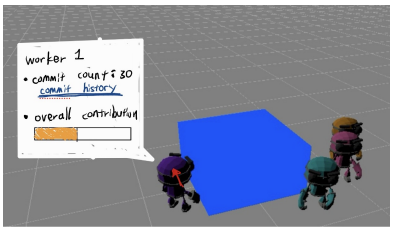


(b) Repository

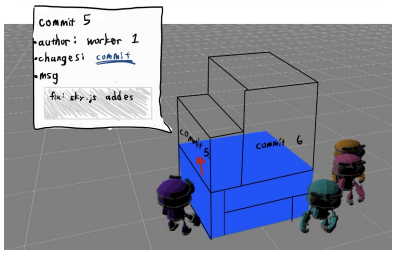


(c) Activity

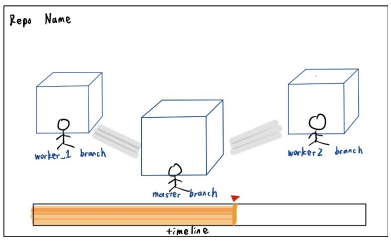
Figure 1: User interface workflow and repository visualization stages.



(a) Worker information



(b) Commit Visualization



(c) Branch Visualization

Figure 2: User interface workflow and repository visualization stages (with additional ideas).

4 Extra Ideas

- Introduce varied worker animations to make the visualization more dynamic.
- Represent branches as separate buildings, turning a repository into a "city" instead of a single structure (2c).
- Allow multiple repositories to be visualized simultaneously (either multiple buildings or multiple cities).
- Use different worker models (not just color variations) to better distinguish contributors. Viewer can select the worker models based on their preference.
- Improve the background and overall visual design for a more engaging experience.
- Implement caching for recently visualized repositories to reduce API calls and save tokens.
- Add a contributor ranking system based on the number of commits.
- Make building bricks interactive, showing the associated commit details when clicked (2b).
- Make the timeline clickable, allowing users to navigate through different points in time and observe how the repository evolves.